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Conceptual design and optimization study of a 3 MW fully superconducting synchronous electrical machine for the electric aircraft based on the magnetic vector potential boundary method

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Keywords: fully superconducting electrical machines, electric aircraft, HTS REBCO windings, magnetic vector potential boundary method, electrical machine optimization

Abstract

Future electrical propulsion systems for electric aircraft need lightweight and compact electrical machines with high power density, high efficiency, and high power-to-weight ratio (PWR). High temperature superconducting (HTS) electrical machine is expected to be a promising technology to meet these challenging demands due to the ultrahigh current density, near-zero DC resistance, light weight, and compact volume of the 2 G HTS REBCO coated conductors. However, in traditional superconducting machine design processes, the unique properties of the HTS machine windings are often oversimplified. These oversimplifications make it challenging to accurately estimate the critical current and loss of the HTS machine windings under the complex background magnetic fields inside the electrical machines. This paper proposes a magnetic vector potential boundary (MVPB) method to design and optimize a 3 MW fully HTS synchronous electrical machine (FHTS-SM). The results show that the MVPB method can consider the superconducting properties in the design and optimization of the electrical machine, and it has highly accurate results. Meanwhile, the MVPB method significantly reduces the quantity and complexity of computations, and the computation time is reduced by 96%. Through a joint simulation (model-FRONTIER & MATLAB & COMSOL), multi-objective optimization was conducted on the 3 MW FHTS-SM design to obtain the optimal design. Moreover, the HTS armature winding is optimized by the multifilamentary method to reduce AC loss. The results show that the optimal design can increase the PWR (machine only) by approximately 25% and improve the efficiency by about 8% simultaneously. This work provides a powerful tool to effectively and accurately estimate the loss of the HTS machine windings at a stable operation state. Meanwhile, it offers a good understanding of designing high power-to-weight and high efficiency superconducting electrical machines. Additionally, the simulation results can provide a basis for the R&D of the superconducting electrical machine prototype for future electric aircraft.

1. Introduction

Transportation electrification is one of the key areas to achieve the goal of net-zero emissions [1]. The electric propulsion system of the future electric aircraft can effectively reduce noise, carbon emissions, and environmental pollution [2, 3]. However, for future aircraft, the electrical machines need to be compact, lightweight, and possess an ultrahigh power-to-weight ratio (PWR) (20–50 kW kg⁻¹) [4–10]. This ultrahigh PWR requirement is a huge challenge for traditional electrical machines. To improve the PWR of electrical machines, 2 G high temperature superconducting (HTS) wires are used to replace copper

wires because the HTS REBCO wires are lightweight, compact, and have an ultrahigh current-carrying capability. Therefore, the superconducting electrical machines used HTS wires provide a promising solution to meet the high PWR requirement of future electric aircraft. At present, several industrial aviation companies have invested a lot of funds in the research and development projects of superconducting electrical machines, such as Airbus [11], Airbus & Toshiba [12], NASA [13, 14], Rolls-Royce [15], and MagniX [16].

Currently, lots of partially superconducting electrical machine designs used different topologies are proposed, such as permanent magnet synchronous electrical machine (SM) [17–19], electric SM [20, 21], axial flux electrical machine [7, 22–25], homopolar inductor machine [26–33], flux switching machine [34, 35], field modulated machine [36, 37], and so on. These studies show that, compared to the traditional electrical machine, the PWR of the partially superconducting electrical machine is improved ($\sim 5\text{--}15\text{ kW kg}^{-1}$). However, the very high PWR demand of future electrical machines is still a challenge for partially superconducting electrical machines. To further improve the PWR of superconducting electrical machines, the fully superconducting electrical machine designs employed HTS field windings and armature windings are proposed. The fully superconducting electrical machines have both high electric loading and high magnetic loading. Consequently, the fully superconducting electrical machines can be designed more compact and lighter to achieve higher PWR. Recent advancements in fully superconducting motor designs have good progress in PWR. Wang *et al* [38] proposed a 1 MW superconducting motor with up-down-assembly magnets, achieving a PWR of 11.55 kW kg^{-1} . In a higher power level, Yutaka *et al* [39] developed a 5 MW synchronous superconducting motor design with potential PWR exceeding 16 kW kg^{-1} . Kalsi's team has made significant contributions through multiple prototypes: their 3 MW synchronous motor design [40] attained 28 kW kg^{-1} PWR, while a subsequent 100 kW design [41] successfully completed field coil and half-scale stator coil testing. Meanwhile, as evidenced by Dezhin and Dezhina 5 MW superconducting generator [42] with PWR $>23\text{ kW kg}^{-1}$. Balachandran *et al*'s 2.5 MW motor [43] demonstrates PWR over 25 kW kg^{-1} , and a 10 MW motor of CRUISE [44] achieves a PWR of 40 kW kg^{-1} . These studies indicate that the fully superconducting electrical machine is a promising option to achieve the high PWR (over 20 kW kg^{-1}).

This paper proposes a magnetic vector potential boundary (MVPB) method to design and optimize a 3 MW fully HTS SM (FHTS-SM). The results show that the MVPB method can consider the superconducting properties in the design and optimization of the electrical machine, and this method has high accuracy (the errors of the AC loss and the dynamic loss are less than 2% and 1%, respectively). Meanwhile, the MVPB method significantly reduces the quantity and complexity of computations, and the computation time is reduced by 96%. Consequently, the MVPB method can accurately and quickly estimate losses at a stable operation state through a unidirectional coupled method. Then, Based on a joint simulation method (modelFRONTIER & MATLAB & COMSOL), the initial 3 MW FHTS-SM design is comprehensively optimized, taking into account the space demand, dynamic loss, AC loss, cost, and PWR (machine-only). As a result, the optimal design can reduce the cost of the superconductor by 45% and increase the PWR (machine-only) of the FHTS-SM by approximately 25%. Moreover, the HTS armature winding is optimized by the multifilamentary method to reduce AC loss. It shows that the 4-filamentary armature winding can reduce the AC loss by approximately 83%, and increase the efficiency of the FHTS-SM by 8% approximately. This work provides a powerful tool to effectively and accurately estimate the loss of the HTS machine windings. Meanwhile, it offers a good understanding of designing high power-to-weight and high efficiency superconducting electrical machines. Additionally, the simulation results can provide a basis for the R&D of the superconducting electrical machine prototype for future electric aircraft.

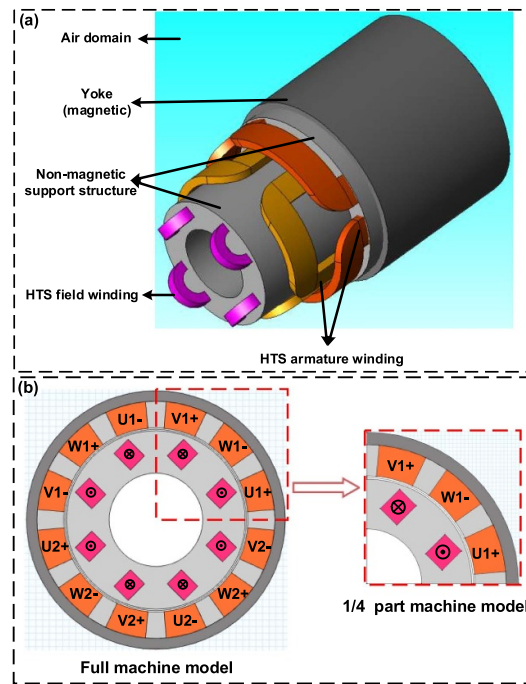
2. Design goal and electrical machine topology

2.1. Design goal of the future electrical machine

For a short-range airplane (Boeing 777), it can carry 300 passengers for 7500 nautical miles (13 890 km) at a cruise speed of Mach 0.84. For security reasons, many engines were distributed on the wingspan to prevent sudden engine failure. Therefore, the average available shaft power (total shaft power/total motor numbers) at cruise speed is the electric power of the motor. Meanwhile, the motor-rated speed is decided by the fan shaft speed, which is calculated by the fan tip speed divided by the fan diameter. Moreover, the working space is also a key constraint for the electrical machine design process. Through these calculations and our survey [4–10, 38–44], the electric power of the HTS motor in the airplane is 3 MW, the rated speed is 5000 rpm, and the motor's stator diameter and axial length should be less than 500 mm. The design requirement of the future electric aircraft motor is shown in table 1.

Table 1. Design goal of the future electrical machine.

Item	—
Electrical power	3000 kW
Rated speed	5000 rpm
Pole-pairs	2
Rated frequency	167 Hz
Stator diameter	<500 mm
Axial length	<500 mm
Line voltage (RMS)	~1000 V
Operation temperature	30 K

**Figure 1.** Topology of the FHTS-SM. (a) 3D structure of the FHTS-SM. (b) 2D structure of the full and 1/4-part FHTS-SM model.

In fully superconducting electrical machines, REBCO tapes are used for both the field windings and armature windings. A 2-pole pair structure is chosen to reduce the machine volume and armature current frequency (167 Hz) and therefore reduce the AC loss in the HTS armature windings. Furthermore, the REBCO windings operate at an LH_2 cryogenic fuel environment (30 K). In this low-temperature environment, the REBCO windings can carry high current (kA scale), and the LH_2 is also heated to a suitable temperature as fuel for the aircraft.

2.2. Topology and modeling method of the FHTS-SM

Figure 1(a) shows the 3D schematic structural diagram of the fully HTS synchronous motor. The FHTS-SM is composed of several key components: the rotor support structure, HTS field windings, HTS armature windings, and the stator support structure. Compared with the traditional synchronous motor, the field copper windings and the armature copper windings are both replaced by HTS REBCO windings in FHTS-SM. Meanwhile, to improve the PWR of the FHTS-SM, the iron core of the rotor and the iron teeth of the stator are all replaced by a non-magnetic support structure. The HTS field windings and HTS armature windings need to be wound according to the design parameters and then packaged and fixed on the rotor and stator, respectively. A significant design consideration is the fragility of the HTS REBCO tape. The HTS REBCO tape can be damaged if overbent. To avoid this irreparable damage, a centralized arrangement of the HTS armature windings is employed rather than the distributed arrangement. In addition, the HTS armature windings are designed with a saddle coil to fix on the stator more easily and obtain well magnetic field distribution. The saddle HTS windings are proposed in [6, 45, 46], and have been successfully manufactured in [46].

To reduce the computation time, the FHTS-SM is simplified to a 2D partial machine model, as shown in figure 1(b). For the electrical machine model in this paper, the material of the back yoke is 1J22 ferro cobalt, and the support structure of the rotor and stator is aluminum. This design can increase the magnetic load of the stator yoke and reduce the total mass. It also shows there are 4 HTS field windings and 6 HTS armature windings in an FHTS-SM. The HTS field windings and armature windings need to be wound according to the design parameters and then fixed to the support structure of the stator and rotor, respectively. Meanwhile, the DC current direction and the three-phase AC current settings are shown in figure 1(b).

3. MVPB method

To improve the reliability of the superconducting electrical machine designs, there are several key demands for the superconducting electrical machine finite element models: (1) in the design and optimization process of the superconducting electrical machines, the superconducting properties of the superconducting machine windings should be considered; (2) the losses of the superconducting machine windings need to be estimated accurately to calculate the efficiency of the superconducting electrical machines; (3) the superconducting electrical machine finite element models should have high convergence and fast computation speed to facilitate a large amount of design and optimization steps. To meet these demands above, we propose a MVPB method for superconducting electrical machine models. In this section, the modeling method and result verification of the MVPB method are introduced and discussed. It should be mentioned that the FHTS-SM model in this section is the initial design of a 3 MW FHTS-SM. This initial design will be analyzed and discussed in detail in section 4.1.

A 2D superconducting electrical machine will ignore the end-winding effect and cause an error. To provide a quantitative estimate or reference for the error introduced by the end-winding effect, we separately analyzed the 2D and 3D errors of the electrical machine model and the superconducting model. According to current research on the error between 2D and 3D models [47–49], the error between the 2D and 3D models is less than 15%. Meanwhile, we compare the total AC loss error between the 2D and 3D HTS winding by simulation which can be found in appendix, and the error is less than 5%. Therefore, the error between 3D and 2D superconducting electrical machines is roughly estimated to be less than 20%. This error of the 2D model is acceptable, and therefore the superconducting electrical machine is designed and optimized by using a 2D model in this paper.

For the design and optimization of the 3 MW fully superconducting electrical machine based on the MVPB method, we assume three key working conditions: (1) In order to design and optimize superconducting electrical machines quickly while considering their superconducting properties, our analysis has ignored the end-effects. (2) The main focus of this work is designing and optimizing the superconducting electrical machine under the rated working conditions. Other working conditions have not been taken into consideration. (3) The cooling equipment can quickly remove the heat generated by the HTS windings, and the operating temperature of the HTS windings is stabilized at 30 K (liquid Hydrogen environment).

3.1. High temperature superconductor modeling

In this paper, the HTS machine windings are built in the HTS machine windings' block based on the T - A formulation. For the T - A formulation model [50–53], the current density of the superconducting layer is calculated by equation (1):

$$\nabla \times T = J \quad (1)$$

where T is the current vector potential of each superconducting layer element, and J is the current density of each superconducting layer element.

For the 2D T - A formulation model, the superconducting layer is simplified to a thin film structure. The current vector potential $T = [T_x, T_y]^T$. Therefore, equation (1) can be converted to equation (2):

$$J_z = \frac{\partial T_y}{\partial x} - \frac{\partial T_x}{\partial y}. \quad (2)$$

For REBCO coated conductors, the relationship between the electric field and current density is presented by the E - J power law [54–62]:

$$E = E_0 \left(\frac{J_{\text{norm}}}{J_c(B)} \right)^n \frac{J}{J_c} \quad (3)$$

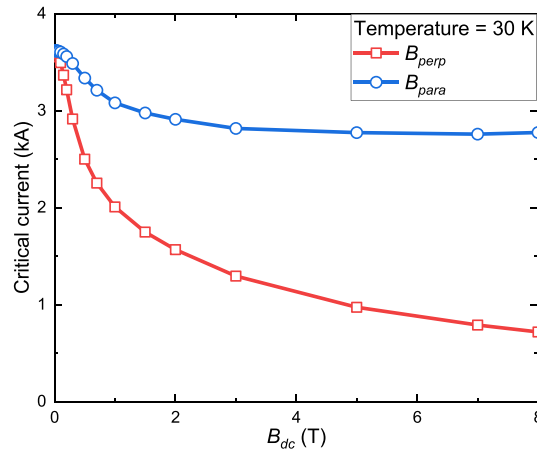


Figure 2. The measuring critical current of the 10 mm Shanghai Superconductor REBCO tape at 30 K from the Robinson HTS wire critical current database [63].

where $E_0 = 10^{-4} \text{ V} \cdot \text{m}^{-1}$ and $n = 28$ at 30 K from the database [63]. B is the magnitude of the magnetic field. In this model, $J_c(B)$ represents the critical current magnetic field dependency function, as shown in equation (4). The operating temperature of the HTS windings is set as 30 K. The impact of the parallel and perpendicular magnetic fields on the critical current of the HTS tape is shown in figure 2,

$$J_c(B) = \frac{I_c(B_{para}) \cdot I_c(B_{perp})}{I_{c0} \cdot w \cdot h} \quad (4)$$

where I_{c0} represents the self-field critical current, and $I_c(B_{para})$ and $I_c(B_{perp})$ represent the impact of the parallel and perpendicular magnetic field on the critical current, respectively. In this model, the magnetic field surrounding the HTS windings can be decomposed into the perpendicular and parallel components with respect to the width direction of the HTS windings by using the ‘vector transformation’ in COMSOL. $I_c(B_{para})$ and $I_c(B_{perp})$ are introduced by the interpolation function from the database [63]. The w and h are the width and height of the thin film HTS tape, respectively.

According to Faraday’s law, we obtain:

$$\nabla \times A = B \quad (5)$$

where A represents the magnetic vector potential of the air domain, and B represents the magnetic flux density of the air domain.

For the 2D model, the governing equation of magnetic vector potential A is:

$$\nabla \times (\nabla \times A_z) = \mu_0 J_z \quad (6)$$

where A_z is the magnetic vector potential. Through equations (2)–(6), the current vector potential T and magnetic vector potential A can be converted and iterated in FEM.

3.2. MVPB modeling method

The oversimplification of the superconducting properties makes it challenging to accurately estimate the critical current and losses of the HTS machine windings under the complex background magnetic fields inside the electrical machines. In order to accurately predict the critical current and losses of the HTS machine windings inside the electrical machine, the FEA models of the superconducting machines must be developed, which couple the superconducting modules and electrical machine modules. One typical approach is to build a fully coupled model combining the superconducting and electrical machine modules [64–73], as shown in figure 3(a). Currently, the fully coupled model based on the T – A formulation is widely used to analyze the electromagnetic characteristics of the superconducting electrical machines in numerous studies [67–73]. However, the fully coupled model often faces two main problems: (1) the nonlinear resistivity of the superconductor. (2) The huge difference in the model size between the superconducting module (1 μm) and the electrical machine module (hundreds of mm). These problems will incur high calculation difficulty, poor convergence, and long computation time

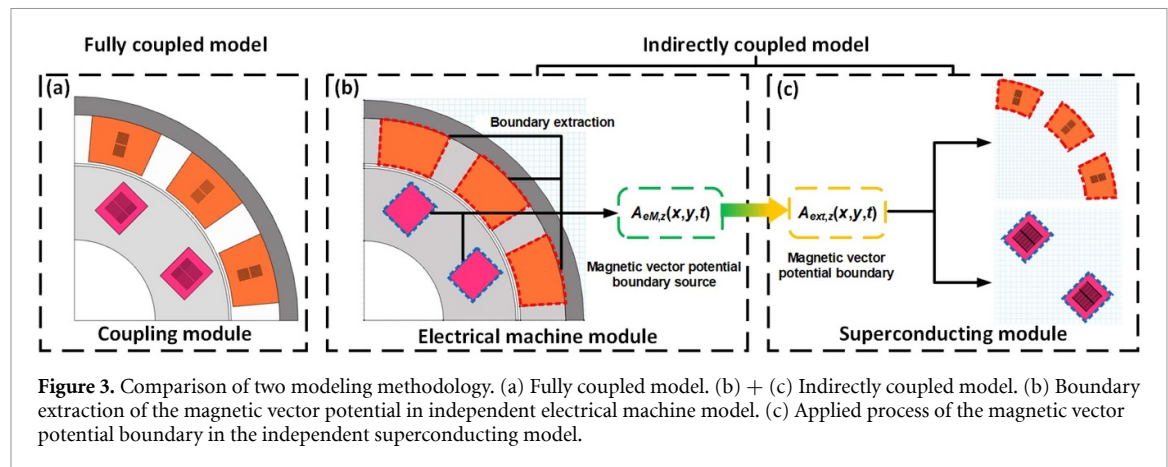


Figure 3. Comparison of two modeling methodology. (a) Fully coupled model. (b) + (c) Indirectly coupled model. (b) Boundary extraction of the magnetic vector potential in independent electrical machine model. (c) Applied process of the magnetic vector potential boundary in the independent superconducting model.

for the fully coupled models. Our recent work [74] proposed an effective solution/approach called the MVPB method. This method can decouple the fully coupled model into independent superconducting modules and an independent electrical machine module, as shown in figures 3(b) and (c). These two modules are indirectly coupled through the unidirectional transmission of the MVPB. In the electrical machine module, the magnetic vector potential surrounding the HTS machine windings is calculated and saved as the MVPB source. This MVPB boundary source is a transient set related to the position. For the T - A formulation superconducting model, the MVPB source is applied to the HTS machine windings' boundaries through the magnetic potential condition to simulate the unique operating condition in the electrical machine. By using this MVPB method, the electromagnetic characteristics of the HTS machine windings can be calculated and analyzed really and accurately.

It is mentioned that the area, mesh, calculated time step, and the relative tolerance of the HTS machine windings are consistent in both the electrical machine model and superconducting model, as shown in figures 3(b) and (c). Meanwhile, we suggest that the width and height of the boundary box need to be at least 15% larger than the size of the superconducting winding. The size of the boundary box mainly affects the convergence of the model. When the size of the boundary box is more than 15% larger than the superconducting winding, its impact on accuracy is negligible. In this paper, the boundary box is more than 15% larger than the superconducting winding.

3.3. Verification of the MVPB method

To verify the MVPB method, the electromagnetic characteristics of the 3 MW FHTS-SM initial design are analyzed separately by the indirectly coupled model and the fully coupled model. The parameters of the electrical machine and superconducting windings are detailed discussed in section 4.2, and the mesh setting of the electrical machine and superconducting domain is the same for the two models. In this section, the simulation results of the indirectly coupled model and the fully coupled model are compared.

The magnetic field distribution and the current density distribution of the HTS machine windings are compared between the indirectly-coupled model and the fully coupled model in figure 4. It can be seen that there are still some differences between the results of the two methods, which are caused by the one-way coupling in the MVPB method. Nevertheless, whether the HTS field winding or the HTS armature winding, it shows that the magnetic field and the current density distribution of both the indirectly coupled model and the fully coupled model are very similar.

Next, the AC loss of the HTS armature windings and the dynamic loss of the HTS field windings are compared in figure 5. It shows that, in general, the AC loss waveform of the HTS armature windings has a high consistency between the two models, and the inaccuracy is lower than 2%. For the dynamic loss of the HTS field windings, the waveform in a period also shows a high consistency between the two models, and the inaccuracy is lower than 1%. The loss comparison of the HTS machine windings further proves the accuracy of the indirectly coupled model.

Moreover, the computational efficiency of the two models is compared in table 2. For both the indirectly coupled model and fully coupled model, the total calculated time is set as 0.1 s with 100 time steps in a magnetic field period, and the relative tolerance is set as 10^{-3} to keep consistency. It shows that the mesh number of the indirectly coupled model is much less than that of the fully coupled model. In terms of computational time, the indirectly coupled model only needs 388 s to obtain the MVPB source in the electrical machine module, and only takes 530 s to calculate the electromagnetic properties of the

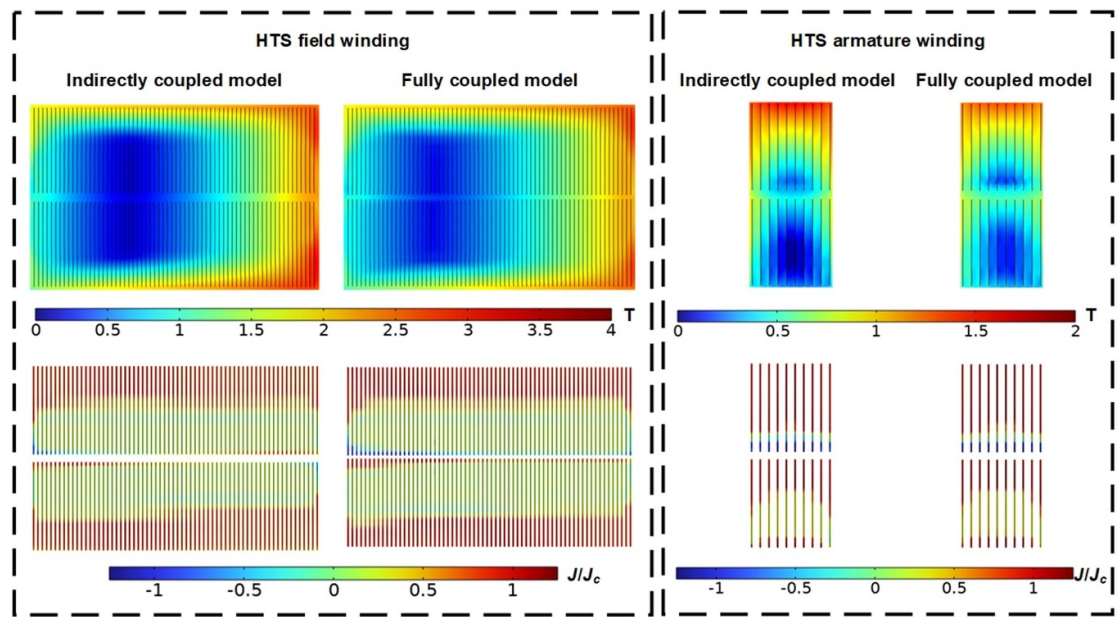


Figure 4. Comparison of the magnetic field and current density distribution of HTS machine windings between indirectly coupled model and fully coupled model.

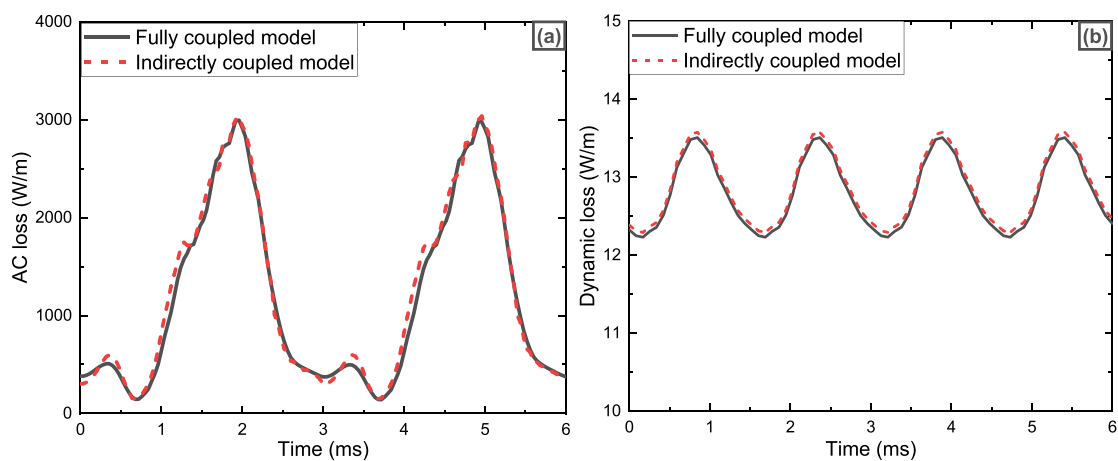


Figure 5. Loss comparison of fully coupled model and indirectly coupled model in a rotational period. (a) AC loss. (b) Dynamic loss.

Table 2. Comparison of the indirectly coupled model and fully coupled model.

Model	Module	Mesh number	Computation time
Indirectly coupled model	EM module	7671	6 min 28 s
	HTS module	5578	8 min 50 s
Fully coupled model	—	27 566	6 h 35 min 25 s

HTS machine windings in the superconducting module. The computation time of the indirectly coupled model consumes 918 s in total. As a comparison, the fully coupled model consumes 23 725 s. Therefore, for the 3 MW FHTS-SM initial model in this paper, the MVPB method can effectively reduce the computation time by 96% compared to the fully coupled model. The results above confirm that the MVPB method can accurately estimate losses at a stable operation state. It should be noted that the MVPB method presented in this paper is a unidirectional coupling method, which is applicable to the stable rated condition analysis and preliminary design of superconducting electrical machines. To analyze the critical scenarios like large load fluctuations, short-circuit faults, or local quenches of superconducting

Table 3. Parameters of initial FHTS-SM design.

Description	Value
Outer diameter of stator (mm)	388
Inner diameter of stator (mm)	276
Air-gap length (mm)	4
Outer diameter of rotor (mm)	268
Inner diameter of rotor (mm)	130
Axial length (mm)	328
Electrical power (kW)	3000
Rated speed (r min^{-1})	5000
Rated phase voltage (V)	1125
Rated phase current (A)	1000
MMF ($\text{kA} \cdot \text{turn}$)	200
Airgap magnetic field (T)	1.3
Pole-pairs	2
Torque (N m)	8383
HTS field windings parameters	Value
Width of the excitation windows (mm)	55
Height of the excitation windows (mm)	25
HTS field windings inner diameter (mm)	50
HTS field windings outer diameter (mm)	100
HTS field windings width (mm)	50
HTS field windings height (mm)	20.5
Number of the HTS field windings	4
Turns of each HTS field windings	200
HTS armature windings parameters	Value
HTS armature windings inner diameter (mm)	110
HTS armature windings outer diameter (mm)	120
Number of the HTS armature windings	6
Turns of each HTS armature windings	20
Mass of each part in FHTS-SM	Value
Stator yoke (kg)	47
HTS armature windings (kg)	15
Stator support structure (kg)	20
Total stator mass (kg)	82
HTS field windings (kg)	25
Rotor support structure (kg)	38
Shaft (kg)	9
Total rotor mass (kg)	72
Total mass (kg)	154
Power to weight ratio (machine only) (kW kg^{-1})	19.5

electrical machines, the MVPB method will be improved to bidirectional coupling for transient analysis in the future.

4. Initial design and optimal design of the FHTS-SM considering the superconducting property

4.1. Initial design of the FHTS-SM

The initial design parameters of the FHTS-SM are shown in table 3. It shows that the excitation magnetomotive force (MMF) needs to reach $200 \text{ kA} \cdot \text{turns}$ to keep the airgap magnetic field value at a high level (1.3 T). The magnetic flux density of the initial design is shown in figure 6, and the maximum of the magnetic flux density is 2.99 T at the HTS field winding. The magnitude of the magnetic field perpendicular to the surface of the HTS field windings is 2.02 T, and the magnitude of the magnetic field parallel to the surface of the HTS field windings is 2.71 T. To meet the current requirements of the 3 MW fully superconducting electrical machine design and provide a margin for current, the

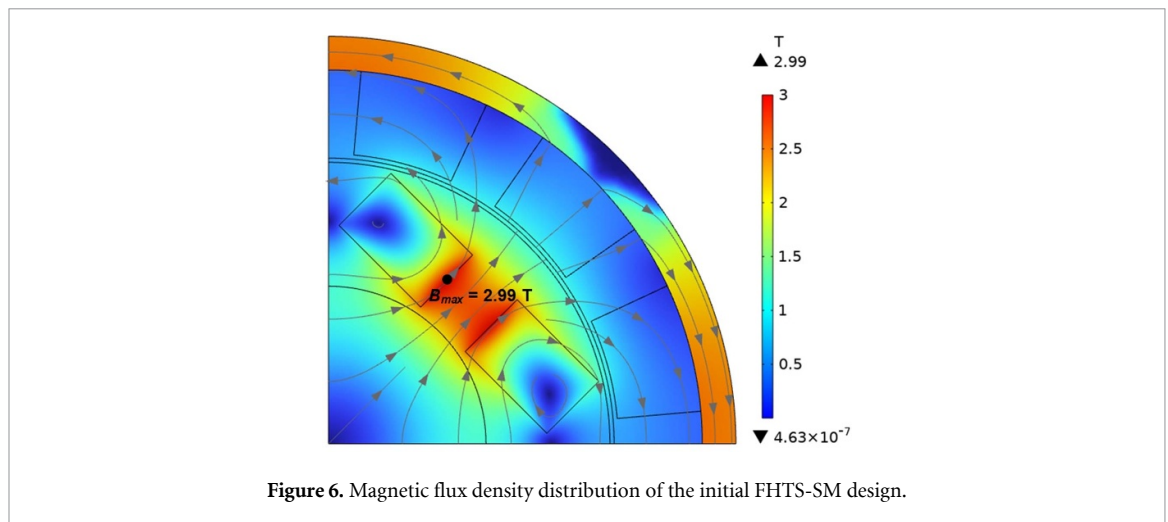


Figure 6. Magnetic flux density distribution of the initial FHTS-SM design.

10 mm-wide REBCO tape is chosen for the HTS machine windings which is manufactured by Shanghai Superconductor. The price of the 10 mm HTS tape is 50 \$ per meter. The operating temperature of the HTS windings is set as 30 K in FHTS-SM. The critical current of the 10 mm HTS REBCO tape under different magnetic fields at 30 K is shown in figure 2 [63]. The self-field critical current of the HTS REBCO tape is 3619 A at 30 K. Meanwhile, the 10 mm HTS REBCO tape consists of a REBCO layer (1 μm), a Hastelloy layer (50 μm), and two Cu layers (20 μm), and the insulation thickness between each turn is 100 μm . Based on the critical current of the REBCO tape as shown in figure 2, the max working DC current of the HTS tape is 1500 A when $B_{\text{prep}} = 2.02$ T. To meet this high MMF, the operating DC current of the HTS field winding is chosen as 1000 A, considering the current safety margin, and one HTS field winding is a 2-coil structure with 100 turns per coil. Meanwhile, the HTS armature windings should also carry three-phase AC current with 1414 A amplitude to ensure the 3 MW output electrical power performance. An HTS armature winding is also a 2-coil structure (standard tape) and 10 turns per coil. The initial design is shown in figure 9, and the width and height of the HTS field winding block are 57 mm and 35 mm, respectively. The average width and height of the HTS armature winding block is 40 mm and 50 mm, respectively. It also shows that the total mass of the FHTS-SM is obviously lighter because of the reduction of the iron core, and therefore, the PWR of the FHTS-SM can reach 19.5 kW kg⁻¹. Notably, this paper mainly focuses on the design and optimization of the parameters of the superconducting electrical machine structure and the superconducting machine windings. The design of the cooling system has not been taken into consideration, and therefore the PWR of the superconducting electrical machine designs is not considered the mass of the cooling systems.

4.2. Optimizing process of the FHTS-SM considering the superconducting properties

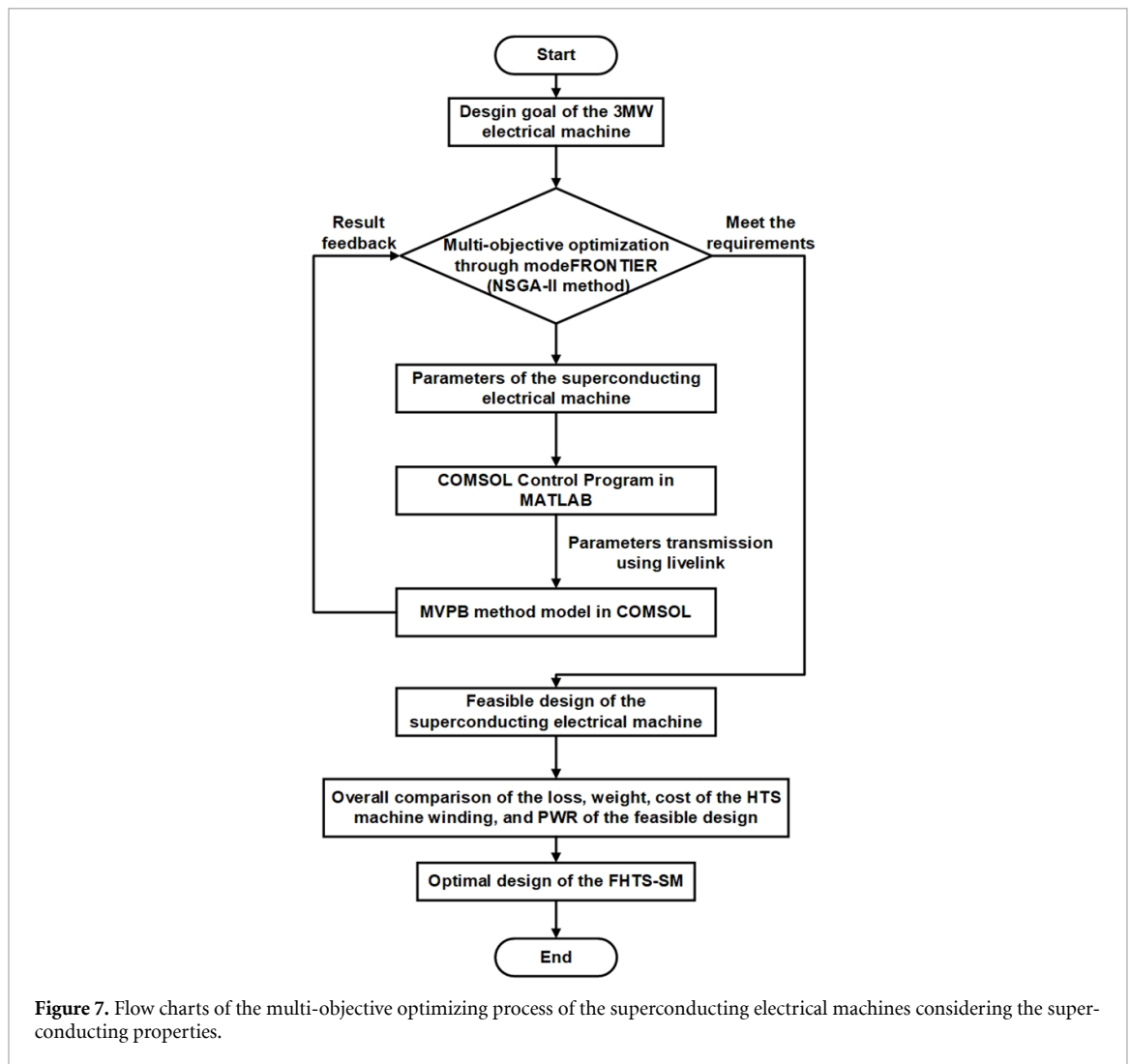
The HTS machine windings in superconducting electrical machines are exposed to the complex magnetic field environment. The critical current of the HTS windings will reduce under this magnetic field. Meanwhile, the HTS armature windings need to carry high AC current, and the HTS field windings need to carry high DC current. Under this working condition, the HTS armature windings will generate large AC loss, and the HTS field windings will generate dynamic loss. It will affect the safety and efficiency of the FHTS-SM. Therefore, to ensure the working stability and safety of the FHTS-SM, it is necessary to consider the superconducting properties of the HTS machine windings in the design and optimization process of the FHTS-SM.

4.2.1. Multi-objective optimizing process of the FHTS-SM

In this part, to achieve the global optimization of the superconducting electrical machines, we combined the global optimization method of traditional motors with the full superconducting motor model presented in this paper. A joint simulation model for superconducting electrical machines parameter optimization was established using modeFRONTIER, MATLAB, and COMSOL.

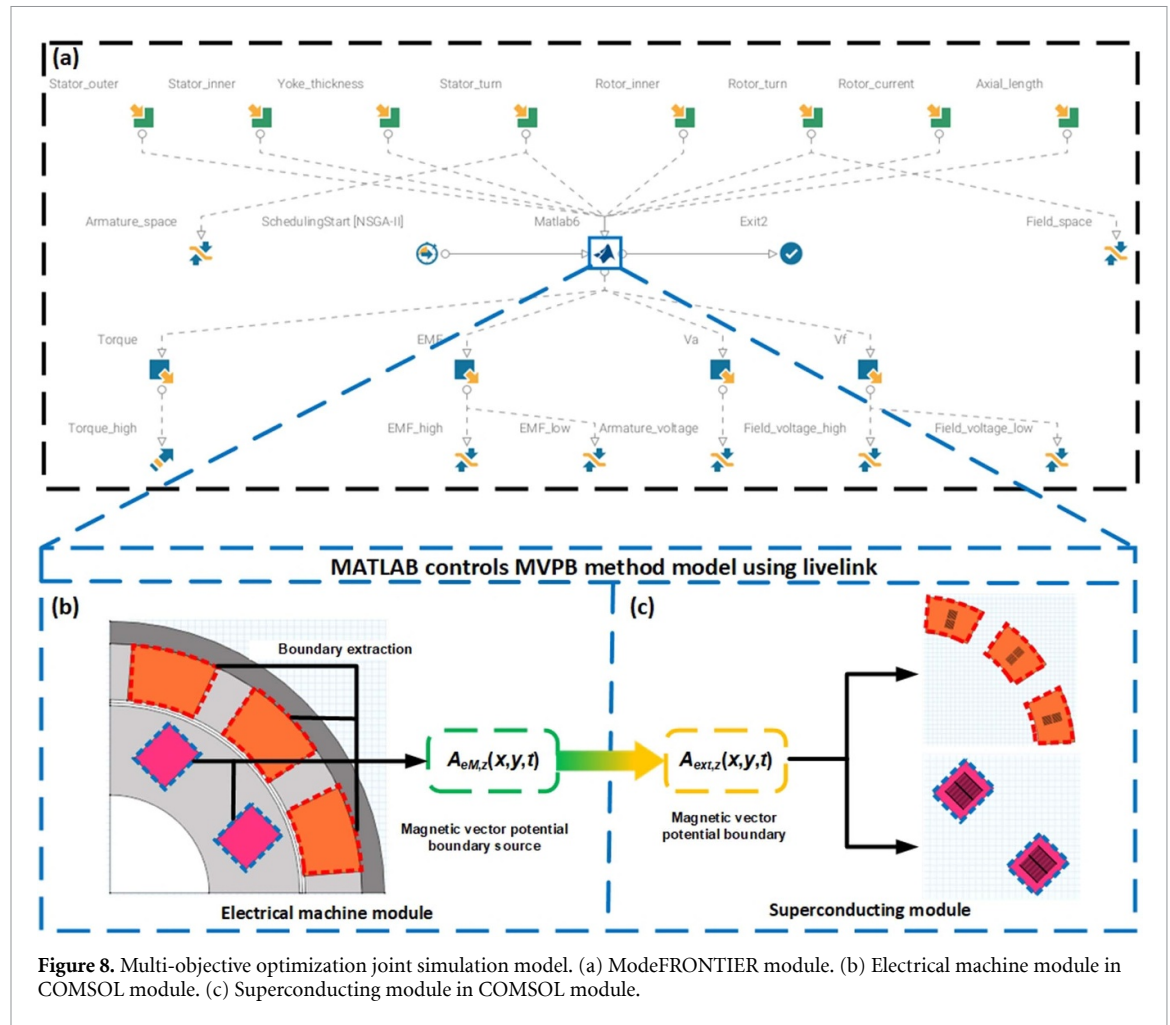
Figure 7 shows the design and optimization process of the superconducting electrical machines, and the main steps are as follows:

- (1) Based on the design goal of the 3 MW fully superconducting electrical machine, the required optimization parameters, objective, and constraint conditions are provided as the basic settings for the modeFRONTIER.



- (2) In the modeFRONTIER module, the superconducting electrical machine parameters are optimized by the NSGA-II method. And these parameters are transmitted as the input parameters of the MATLAB module, such as the outer diameter of the stator, inner diameter of the stator, inner diameter of the rotor, turn number of HTS armature winding, turn number of HTS field winding, and so on.
- (3) In a multi-objective optimization joint simulation model, the MATLAB module is used as the parameter transfer station because modeFRONTIER is unable to directly control COMSOL. In MATLAB, the program is capable of automatically transmitting the electrical machine parameters, opening, and closing the COMSOL model through the COMSOL-MATLAB livelink interface.
- (4) In the COMSOL module, the superconducting electrical machine model will be built and simulated based on the electrical machine parameters from MATLAB. Meanwhile, the results of each superconducting electrical machine are saved and transmitted to the modeFRONTIER module, such as the voltage of the HTS field winding, the voltage of the HTS armature winding, the back electromotive force of the electrical machine, and the torque.
- (5) When the modeFRONTIER receives the result feedback, the modeFRONTIER will determine whether this design meets the objective and constraint conditions. If it is feasible, it will be stored as a feasible design. Meanwhile, these results will also serve as the feedback for NSGA-II, which is used to adjust the input parameters of the next electrical machine design and go back to step (2).
- (6) When all superconducting electrical machine designs are completely calculated, all feasible designs of the superconducting electrical machines are overall compared to obtain the optimal design.

Based on this design and optimizing method of the superconducting electrical machines, the initial 3 MW FHTS-SM design is comprehensively optimized based on the MVPB method in the following sections, taking into account the space demand, dynamic loss, AC loss, cost, and PWR. Notably, this



paper mainly focuses on the design and optimization of the parameters of the superconducting electrical machine structure and the superconducting machine windings. The design of the cooling system has not been taken into consideration, and therefore the PWR of the superconducting electrical machine designs is not considered the mass of the cooling systems.

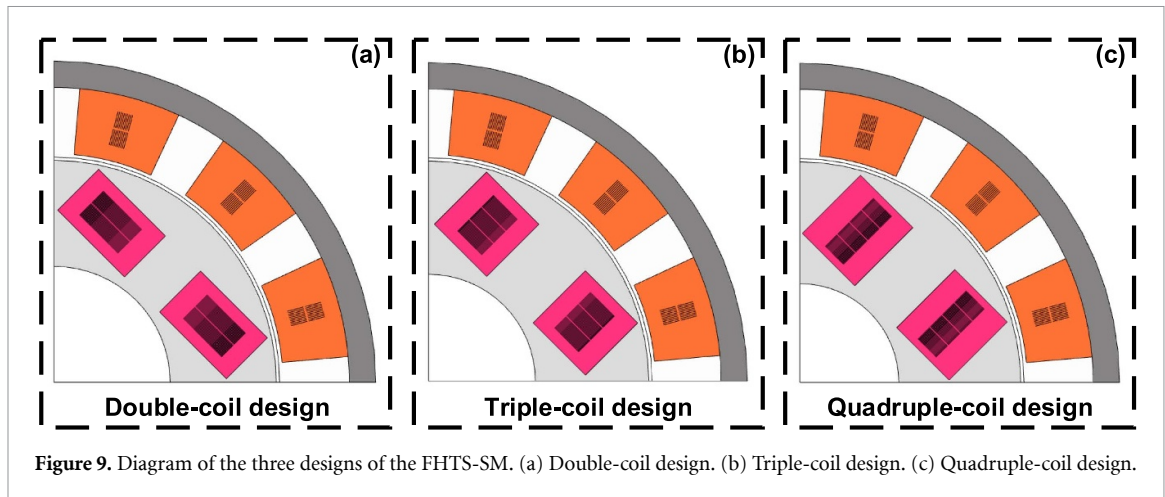
4.2.2. Multi-objective optimization joint simulation model of the FHTS-SM

In this part, the settings of the multi-objective optimization joint simulation model are introduced. The specific settings of the modeFRONTIER model are shown in figure 8(a). There are 8 input variables of superconducting electrical machines:

- (1) Stator outer diameter.
- (2) Yoke thickness.
- (3) Stator inner diameter.
- (4) Number of HTS armature winding turns.
- (5) Rotor inner diameter.
- (6) Number of field winding turns.
- (7) Amplitude of the field winding current.
- (8) Axial length.

These input variables are optimized by the NSGA-II method. As shown in figure 8(a), there are 8 objective and constraint conditions:

- (1) Space of the HTS armature windings.
- (2) Space of the HTS field windings.
- (3) Maximum of the EMF (RMS)—1200 V.
- (4) Minimum of the EMF (RMS)—1000 V.
- (5) Voltage criterion of the HTS armature winding—lower than $1 \mu\text{V cm}^{-1}$.



- (6) Voltage criterion of the HTS field winding—lower than $1 \mu\text{V cm}^{-1}$.
- (7) Voltage criterion of the HTS armature winding—higher than $0.95 \mu\text{V cm}^{-1}$.
- (8) Maximum of the torque.

Where, condition (1) and (2) are used to constrain the space of the HTS armature and field windings. In this paper, the gap between each turn of the HTS machine winding is set as 0.5 mm, and the gap between each coil is set as 0.5 mm. Then, the width, height, and area of the HTS armature and field windings can be calculated. Conditions (3) and (4) are for controlling the EMF (RMS) to be approximately equal to 1000 V. Conditions 6 and 7 are designed to find the critical current of the HTS field winding. It should be noted that two electrical machine modules need to be built in COMSOL. The first electrical machine module and superconducting module is used to obtain the voltage of the HTS armature/field winding and calculate the critical current. In the second electrical machine module, the DC current of the HTS field winding needs to be set at 70% of that of the HTS field windings in the superconducting module to meet the requirement of leaving a 30% current margin [75] and obtain EMF and torque. Taking these conditions as the optimization targets, a feasible solution is ultimately obtained.

Therefore, based on the same optimization objective, we conducted multi-objective optimization for three different field winding arrangement designs (double-coil, triple-coil, and quadruple-coil), respectively. The diagram of the three designs of the FHTS-SM is shown in figure 9. After obtaining the corresponding feasible designs for the three different field winding configurations, we calculated the torque, PWR, and efficiency for each design, and plotted scatter plots as shown in figures 10 and 11. It can be seen that the superconducting electrical machine designs with 3-coil and 4-coil structures have higher efficiency and PWR, while the 2-coil structure superconducting electrical machine design has a higher torque. In order to better select the superconducting electrical machine design, we chose the design with the highest PWR, the highest efficiency, and the highest torque for a comprehensive presentation and comparison. As shown in figures 10 and 11, the highest PWR, highest efficiency, and highest torque designs are marked with black circles. In follow parts, the highest PWR, highest efficiency, and highest torque designs are defined as design 1, design 2, and design 3, respectively.

4.3. Structure parameters and critical current of the HTS machine windings in three feasible FHTS-SM designs

Based on a multi-objective optimization joint simulation model, the structure parameters and critical currents of three feasible designs are obtained. As shown in table 4, the HTS field winding designs of design 1, design 2, and design 3 are $4 * 31$, $3 * 55$, and $2 * 78$, respectively. By comparing the parameters of three HTS field winding designs in table 4, it shows that the design 1 ($4 * 31$) is the optimal design because of the less HTS material, lower weight, and lower cost. Compared to the initial design, the total length and cost of the HTS 4-coil field winding (design 1) can be reduced by 62%. It shows that optimizing the HTS field winding can significantly reduce the cost of the superconducting materials and increase the power density of the HTS field winding.

For HTS armature windings, the HTS armature winding designs of design 1, design 2, and design 3 are $2 * 14$, $2 * 12$, and $2 * 14$, respectively. As shown in table 4, the design 2 has the highest critical current and the lowest total length. However, compared to the initial design, the total length of the HTS armature winding in design 2 increases by 66%. Overall, the reduction in the total length of

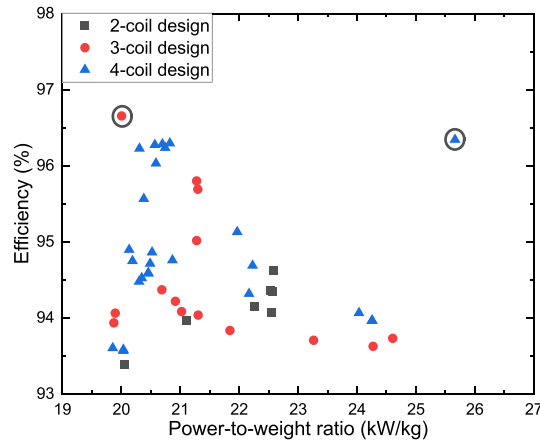


Figure 10. The power-to-weight ratio and efficiency of the feasible 3 MW superconducting electrical machine designs.

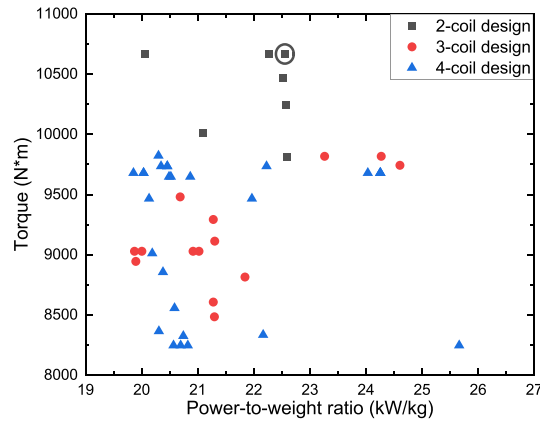


Figure 11. The power-to-weight ratio and torque of the feasible 3 MW superconducting electrical machine designs.

the HTS field winding is significantly greater than that of the HTS armature winding. Therefore, the multi-objective optimization joint simulation method can optimize the weight and economy of the HTS machine windings. The cost of the HTS machine windings is merely used for the economic comparison of different superconducting electrical machine designs and does not serve as the selection criterion for optimizing the design in this paper.

4.4. Loss analysis of the HTS machine windings

In this section, the dynamic loss of the HTS field windings and the AC loss of the HTS armature windings will be analyzed by the indirectly coupled superconducting electrical machine model. The loss q (W m^{-1}) of the HTS machine windings can be calculated by equation (7):

$$q = \frac{1}{N} \sum_{i=1}^N \int_{t_1}^{t_1+T} \int_s E_i \cdot J_i ds dt \quad (7)$$

where N is the number of turns of the HTS machine winding, T is the time of a magnetic field period, s represents the area of the superconducting layer, E_i is the electric field of each turn, and J_i represents the current density of each turn.

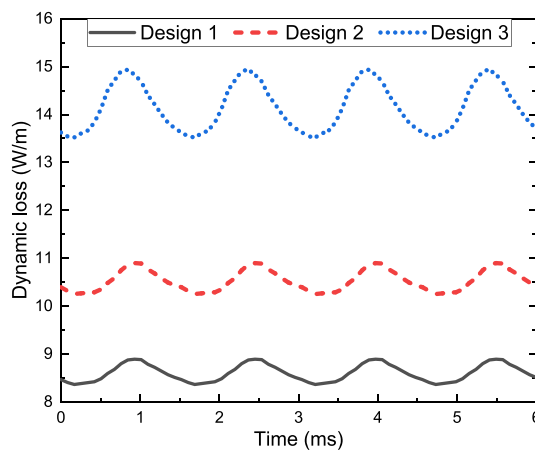
4.4.1. Dynamic loss of the HTS field windings

Figure 12 compares the total dynamic loss of the HTS field windings of three designs. It shows the dynamic loss of design 1, design 2, and design 3 is 8.6 W m^{-1} , 10.7 W m^{-1} , and 14.4 W m^{-1} , respectively. Design 1 (quadruple-coil structure) is the smallest loss scheme of the three designs, which can reduce the dynamic loss by 23%. It can be seen that among the three designs, design 1 has the lowest dynamic loss per meter. Meanwhile, it can be calculated that the total dynamic loss of the HTS field

Table 4. Optimal results of HTS machine winding of three designs.

Design	1	2	3
HTS field winding			
Number of coils	4	3	2
Number of turns	31	55	78
HTS tape width (mm)	10	10	10
REBCO layer thickness (μm)	1	1	1
Critical current of HTS winding (A)	2320	2040	2120
Working current (A)	1624	1428	1484
Current safety margin	30%	30%	30%
Total length (m)	261	343	322
Total weight (kg)	1.47	2.18	2.09
HTS field windings width (mm)	33	21	16
HTS field windings height (mm)	20.5	31	41.5
Cost of the HTS field windings ¹ (k\$)	13.1	17.2	16.1
Power density (MW km^{-1})	11.5	8.7	9.3
HTS armature winding			
Number of coils	2	2	2
Number of turns	14	12	14
HTS tape width (mm)	10	10	10
Critical current of HTS winding (A)	2481	2633	2432
Working current (A)	1414	1414	1414
Current safety margin	43%	46.3%	41.8%
Total length of HTS tape (m)	176	128	178
Total weight (kg)	1.19	0.79	1.21
Cost of HTS armature windings ¹ (k\$)	8.8	6.4	8.9

1-The price of the 10 mm width HTS tape is 50 \$ per meter. The price changes every year, and the cost estimation needs to be calculated based on the actual price of that year. The 4-filament tape is assumed to have the same per-meter cost as the standard HTS tape for estimation; note that this is a projected cost scenario, as current manufacturing maturity is not quite there yet. In addition, the long-length multi-filament tape production may incur a price premium, which will decrease with manufacturing maturity.

**Figure 12.** Comparison of the dynamic loss of the HTS field windings of three FHTS-SM designs in a rotational period.

windings of design 1 is 2.2 kW, accounting for only 0.07% of the total power of the 3 MW FHTS-SM. This is because the AC field component is small at the HTS field windings in FHTS-SM, and the dynamic resistance effect in the HTS field winding is also weakened. These results indicate that the HTS field windings have high performance in FHTS-SM.

To detailly understand the loss characteristics of the HTS field winding, the dynamic loss of each coil of the three designs is shown in figure 13. It shows that the dynamic loss of the bottommost coil is obviously higher than that of the topmost coil for three designs. Among the three designs, the dynamic loss of the bottommost coil in design 3 is the highest, followed by design 2, and design 1 has the lowest. This

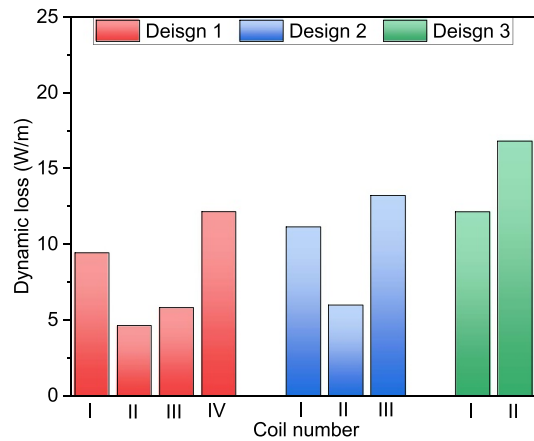


Figure 13. Dynamic loss comparison of each coil in HTS field winding of three designs.

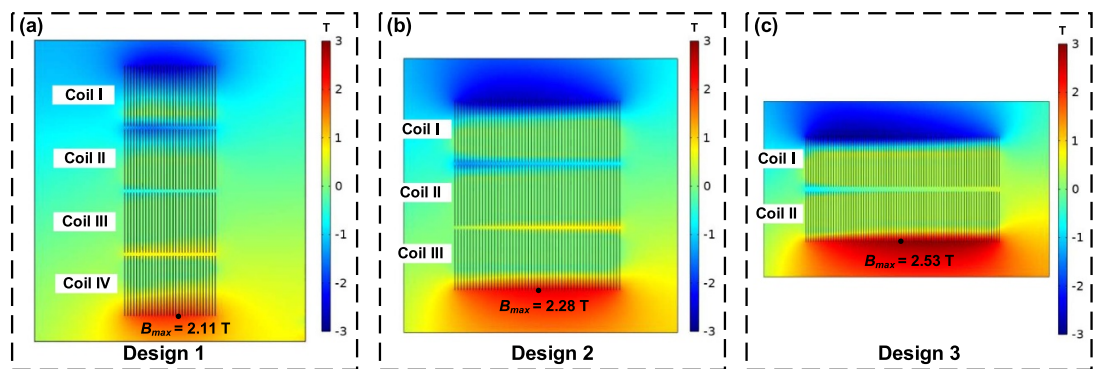


Figure 14. Perpendicular magnetic field distribution of HTS field windings (left part) in three designs at 1/4 of the rotational period. (a) Design 1. (b) Design 2. (c) Design 3.

is because, as shown in figure 14, the perpendicular magnetic field at the bottommost coil of design 1, design 2, and design 3 is 2.11 T, 2.28 T, and 2.53 T, respectively. It also shows that the dynamic loss of the external coils is obviously higher than the dynamic loss of the internal coils for design 1 and design 2. This is because the perpendicular magnetic field at the internal coils is obviously small due to the shielding effect, as shown in figure 14. Therefore, the average dynamic loss of the design 1 and design 2 is lower than that of the design 3. It also relevant that the dynamic losses of the HTS field winding are mainly contributed by the external coils.

4.4.2. AC loss of the HTS armature windings

Figure 15 compares the total AC loss of the HTS armature windings of three designs. It shows the AC loss of design 1, design 2, and design 3 is 607 W m^{-1} , 663 W m^{-1} , and 1083 W m^{-1} , respectively. The lowest AC loss per meter scheme of the HTS armature windings is design 1. However, the lowest total AC loss design is design 2 due to its shortest total length of the HTS armature windings. The total AC loss of the HTS armature windings of design 2 is 85 kW, accounting for 2.9% of the total power of the 3 MW FHTS-SM. Notably, the AC loss of the HTS armature windings is very high because of the high background magnetic field in FHTS-SM. To deeper understand the AC loss characteristics of the HTS armature windings, the AC loss of each coil is compared. Taking the HTS armature windings in phase V as an example, the loss characteristics of each coil in three designs are presented in figure 16. It shows that the AC loss of design 3 is the highest, followed by design 2, and design 1 has the lowest. This is because the perpendicular magnetic field in design 3 is the highest, and that in design 1 is the lowest, as shown in figure 17. Moreover, it also shows that the AC loss of coil II is obviously higher than the AC loss of coil I in the three designs. This is because the position of the coil II is nearly to the airgap of the superconducting electrical machine, so the coil II is exposed to a higher perpendicular AC magnetic field. Consequently, in order to reduce the AC loss, the HTS armature windings need to be installed as far away from the airgap for the air-core fully superconducting electrical machine.

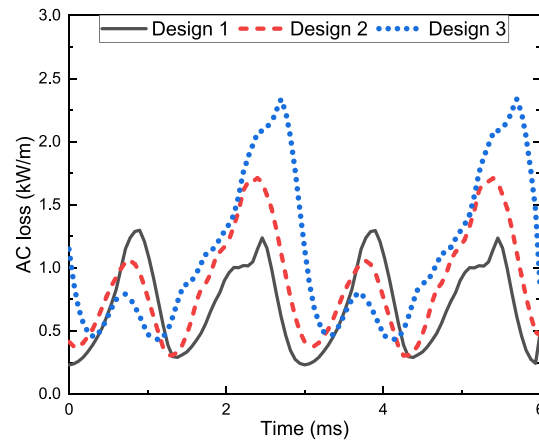


Figure 15. Comparison of the AC loss of the HTS armature windings of three FHTS-SM designs in a rotational period.

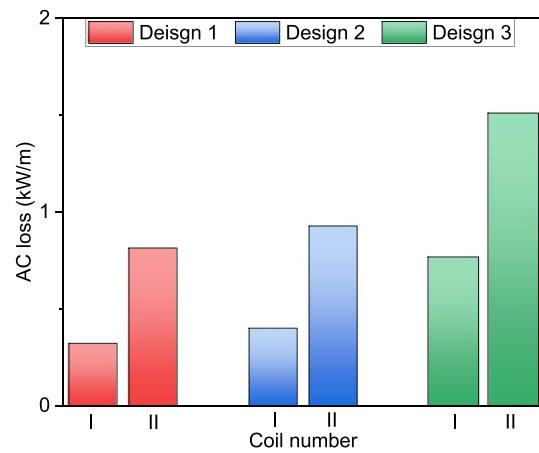


Figure 16. AC loss comparison of each coil in HTS field winding of three designs.

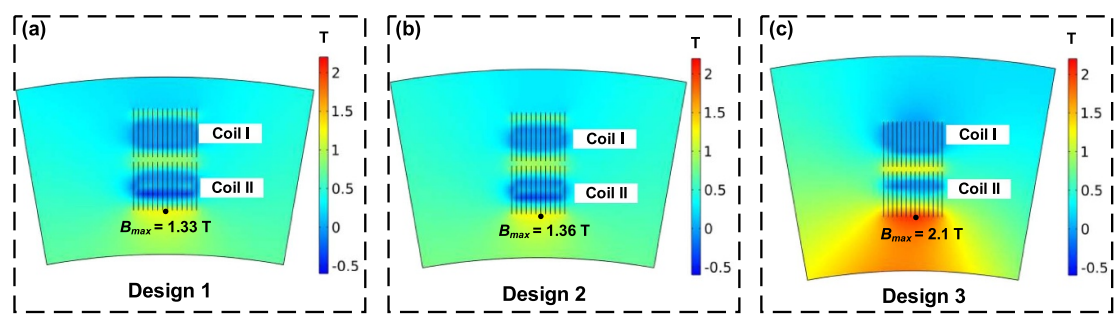
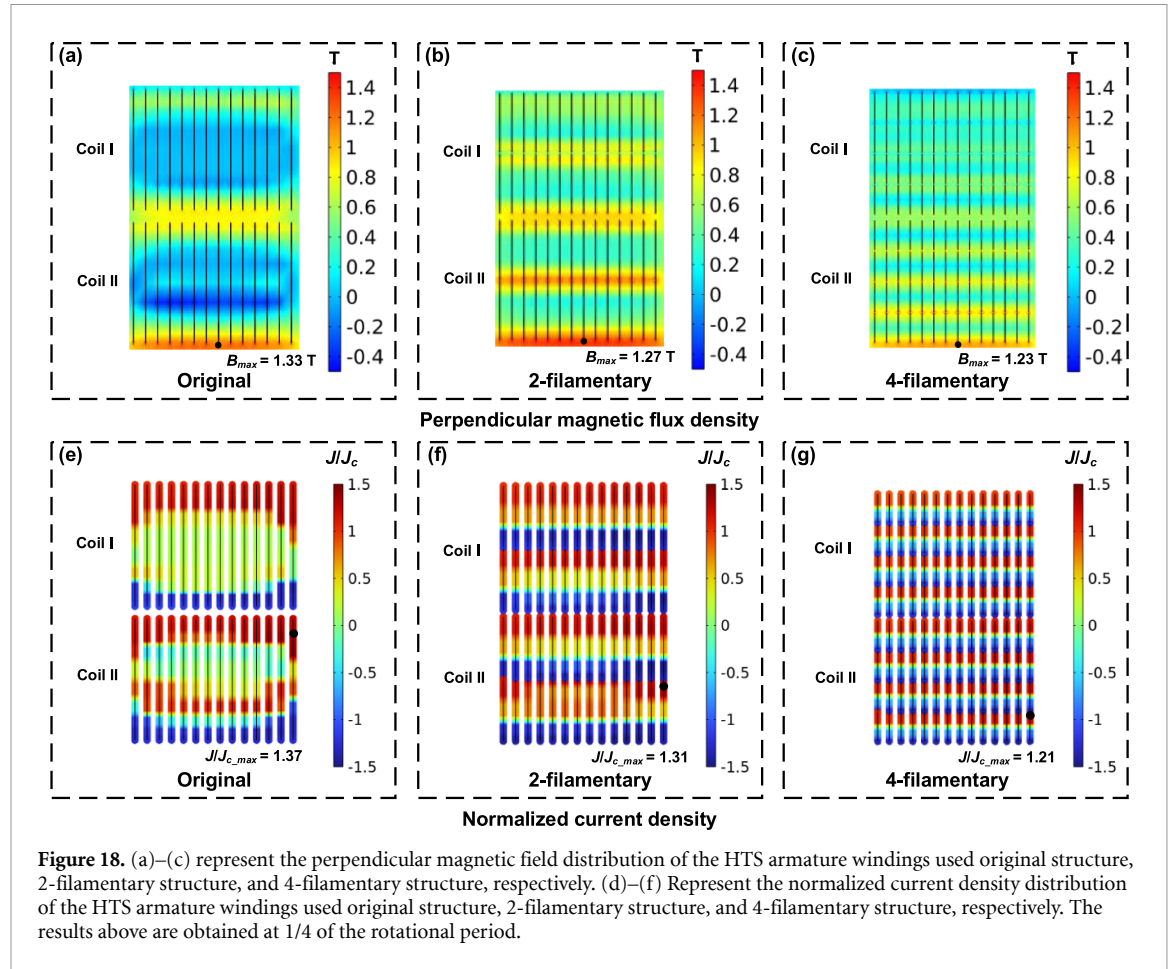


Figure 17. Perpendicular magnetic field distribution of HTS armature windings (phase V) in three designs at 1/4 of the rotational period. (a) Design 1. (b) Design 2. (c) Design 3.

Through the analysis above, it can be found that design 1 has high efficiency and the highest PWR among the three feasible designs. Therefore, design 1 is chosen as the optimal design of the 3 MW fully superconducting electrical machine and further studied in follow sections.

4.4.3. AC loss reducing method of the HTS armature windings

The results above show that the AC loss of the HTS armature windings is high in FHTS-SM, which is 20 times higher than the dynamic loss. To ensure the operating stability and safety of the FHTS-SM, it is necessary to reduce the AC loss of the HTS armature windings. In this section, the multifilamentary method of the HTS armature windings is used to reduce the AC loss. For the HTS filamentary structure



windings, the groove between the filamentary REBCO layer is set as 20 μm . When the filament width is less than 2 mm, the critical current will show a significant reduction. Therefore, in order to maintain the critical current of the HTS tape, the maximum number of filaments is selected as 4. Based on the design 3 of the FHTS-SM, the AC loss of the original structure (10 mm), 2-filamentary structure (each filament is 5 mm), and 4-filamentary structure (each filament is 2.5 mm) of the HTS armature windings are analyzed and compared.

Figures 18 (a)–(c) show the perpendicular magnetic field distribution of the HTS armature windings used original structure, 2 filaments structure, and 4 filaments structure, respectively. It shows that the perpendicular magnetic field of the HTS armature windings gradually decreases with the increase of the filament number. The maximum magnetic field of these three cases is 1.33 T, 1.27 T, and 1.23 T. This is because the shielding effect of the HTS armature windings is weakened with the increase of the filaments, and therefore the perpendicular magnetic field can penetrate easily the HTS armature windings with filamentary structure, leading to the decrease of the maximum of the magnetic field at the two ends of the HTS armature windings. Figures 18 (d)–(f) show the normalized current density distribution of the HTS armature windings used original structure, 2 filaments structure, and 4 filaments structure, respectively. It shows that the current density of the HTS armature windings gradually decreases with the increase of the filament number. The maximum normalized current density of these three cases is 1.37, 1.31, and 1.21. This is because the HTS armature windings are penetrated easily with the increase of the filaments, leading to a weaker skin effect. Therefore, the AC transport current region expands, and the maximum of the current density decreases. In addition, the shielding current decreases due to the weaker shielding effect and lower magnetic field which also reduces the current density of the HTS armature windings. In summary, the multifilamentary method can effectively reduce the perpendicular magnetic field and the current density of the HTS armature windings.

Figure 19 compares the AC loss of the HTS armature windings using original structure, 2 filaments structure, and 4 filaments structure. It shows that the AC losses of these three cases are 607 W m^{-1} , 483 W m^{-1} , and 270 W m^{-1} , respectively. Theoretically, the AC losses of the HTS armature windings with 2-filamentary structure and 4-filamentary structure should be half and a quarter of those of the original structure, respectively. But the AC losses of the HTS armature windings with 2-filamentary

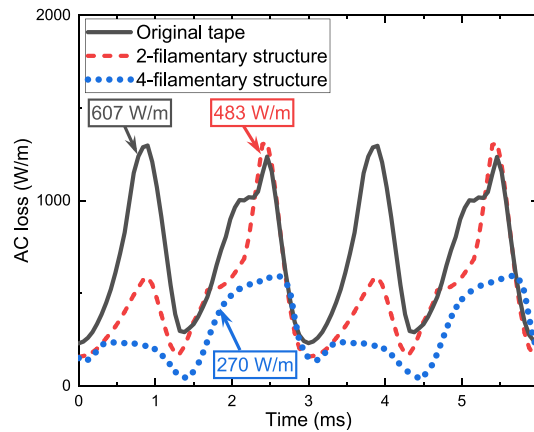


Figure 19. Comparison of the AC loss of different HTS filamentary windings in a rotational period.

structure and 4-filamentary structure is higher than the theoretical results. This is because the inter-filament coupling currents of the multifilamentary structure produce extra coupling loss. It also shows that the AC loss is obviously reduced with the increase of the filament number because of the lower perpendicular magnetic field and lower current density. When the 4 filamentary structure is used, the AC loss of the HTS armature windings is reduced by 56%. Consequently, the HTS armature windings in 3 MW FHTS-SM adopt the 4 filamentary structure to reduce the AC loss. In addition, the HTS armature winding should be designed with the narrowest possible width to minimize the AC losses for future fully HTS electrical machines.

The AC loss of the HTS armature windings using a 4-filamentary structure is 47.5 kW in our design. We believe that the AC loss generated by the HTS windings in a 3 MW fully HTS superconducting electrical machine can be taken away in future electric aircraft. This is because future electric aircraft use liquid hydrogen as the main fuel. Then it can be revealed that the heat required to convert the liquid hydrogen coolant in the superconducting electrical machine into gaseous form will be higher than the amount of cold power needed by HTS armature windings [6]. Thus, the cooling comes almost ‘for free’ and ‘useful’. This means not only that cooling technology for HTS armature windings is only a small burden on e-aircraft, but that higher cryogenic loss in HTS armature windings can be tolerated.

4.5. Overall comparison of the initial design and optimal design

Through the optimization of the electrical machines considering the superconducting properties, the optimal 3 MW FHTS-SM design is obtained. The parameters of the optimal design are shown in table 5, and the parameters of the initial design are also given in table 5 as a comparison. The detailed performance of the 3 MW FHTS-SM is shown in table 3. It should be noted that the multi-filamentary tapes will increase the cost of the HTS armature windings. However, the customized price of HTS multi-filamentary tape depends on the specific design requirements, and meanwhile the price of manufacturing long-length multi-filamentary HTS tape is uncertain. Therefore, the price of the original 10 mm HTS tape is used for the cost estimation of multifilamentary tape in this paper. Noted that this is a projected cost scenario, as current manufacturing maturity is not quite there yet. It shows that, compared with the initial design, the optimal design can obviously reduce the cost of the HTS machine windings by 45% in total. In terms of the losses of the HTS machine windings, the HTS field windings used a quadruple-coil structure can reduce the dynamic loss by 75.3%, and the AC loss of the HTS armature windings adopting the 4 filamentary structure is reduced by 83.3%. The efficiency of the optimal design can be increased by 8.12%. Moreover, the optimal design can also reduce the total weight of the FHTS-SM by 25.3%, and the PWR of the FHTS-SM is increased by 25.3%. Notably, the PWR of the fully superconducting electrical machine does not take into account the weight of the cooling system in this paper. The results above demonstrate the necessity of considering the superconducting properties of the HTS machine windings in the design and optimization process, which enhances the economic and reliability of superconducting electrical machine designs.

Table 5. Comparison of initial design and optimal design.

FHTS-SM parameters	Initial design	Optimal design
Outer diameter of stator (mm)	388	380
Inner diameter of stator (mm)	276	281
Air-gap length (mm)	4	4
Outer diameter of rotor (mm)	268	273
Inner diameter of rotor (mm)	120	127
Axial length (mm)	328	325
Rated electrical power (kW)	3375	3033
Rated speed (r min ⁻¹)	5000	5000
Phase voltage (RMS) (V)	1125	1011
Phase current (RMS) (A)	1000	1000
MMF (kA * turn)	200	189
Airgap magnetic field (T)	1.3	1.1
Pole-pairs	2	2
HTS field winding details	Initial design	Optimal design
Operating DC current (A)	1000	1484
Winding width (mm)	50	16
Winding height (mm)	20.5	41.5
Winding arrangement	2-coil	4-coil
Tape structure	Original	Original
Number of the windings	4	4
Turns of each winding	200	124
Total length (m)	692	262
Cost ¹ (k\$)	34.6	13.1
Dynamic loss (kW)	8.9	2.2
Dynamic loss ratio	0.29%	0.07%
HTS armature windings	Initial design	Optimal design
Operating AC current (A)	1414	1414
Winding width (mm)	5	5
Winding height (mm)	20.5	20.5
Winding arrangement	2-coil	2-coil
Tape structure	Original	4-filament
Number of the windings	6	6
Turns of each winding	20	28
Total length (m)	106	177
Cost ¹ (k\$)	5.3	8.85
AC loss (kW)	285	47.5
AC loss ratio	9.5%	1.59%
Mass of each part	Initial design	Optimal design
Stator yoke (kg)	47	37.5
HTS armature windings (kg)	15	21.5
Stator support structure (kg)	20	18
Total stator mass (kg)	82	77
HTS field windings (kg)	35	12
Rotor support structure (kg)	28	19
Shaft (kg)	9	7
Total rotor mass (kg)	72	38
Total mass (kg)	154	115
Power to weight ratio ² (machine only) (Kw kg ⁻¹)	19.5	26.1
Efficiency	90.21%	98.34%

1-The price of the 10 mm width HTS tape is 50 \$ per meter. The price changes every year, and the cost estimation needs to be calculated based on the actual price of that year. The 4-filament tape is assumed to have the same per-meter cost as the standard HTS tape for estimation; note that this is a projected cost scenario, as current manufacturing maturity is not quite there yet. In addition, the long-length multi-filament tape production may incur a price premium, which will decrease with manufacturing maturity.

2-The power-to-weight ratio of the fully superconducting electrical machine does not take into account the weight of the cooling system in this paper. The PWR result will be somewhat overestimated.

Table 6. Parameters of the 10 kW fully superconducting electrical machine and REBCO armature winding.

Description	Value
Outer diameter of stator (mm)	154
Inner diameter of stator (mm)	102
Air-gap length (mm)	1
Outer diameter of rotor (mm)	100
Inner diameter of rotor (mm)	40
Axial length (mm)	100
Electrical power (kW)	10
Rated speed (r min^{-1})	5000
Rated phase voltage (V)	150
Rated phase current (A)	30
Pole-pairs	2
HTS armature windings parameters	Value
Tape width (mm)	4
REBCO layer thickness (μm)	1
Tape self-field I_{c0} at 77 K (A)	223
Coil inner diameter (mm)	44
Coil outer diameter (mm)	56
Straight part length of coil (mm)	100
Number of coils	2
Turn number of each coil	12
Total length of REBCO winding (m)	8.9
Tape self-field I_{c0} at 77 K (A)	143

5. AC loss experiment of a REBCO winding for the 10 kW fully superconducting electrical machine

The manufacturing of the MW-scale superconducting electrical machines requires advanced manufacturing techniques and substantial financial support. Therefore, to validate the superconducting electrical machine design method, we have designed a small-scale (10 kW) fully superconducting SM based on the design method of superconducting electrical machines in this paper. The parameters of the 10 kW fully superconducting electrical machine, REBCO armature winding, and REBCO tape are listed in table 6.

Based on the design in table 6, we wound superconducting racetrack-shaped windings of the same size using 4 mm superconducting tape from Shanghai Superconducting Company, as shown in figure 20(a). The superconducting armature winding is wound on a racetrack-shaped former, which has a double-coil structure with 12 turns on each of the upper and lower coils. To avoid burrs during the manufacturing process, the former is composed of two parts—the upper and lower sections. After the upper and lower coils of the winding are wound, the upper and lower parts of the winding are fixed using bolts. Meanwhile, to compensate for the phase angle between the measured AC voltage signal and the AC current signal of the HTS REBCO winding, we wound a copper winding of the same size as the HTS REBCO winding, as shown in figure 20(b).

To measure the AC transport losses of the HTS REBCO winding, the AC loss measurement platform is built, as shown in figure 20(c). We used the traditional electrical measurement method (four-probe method) to measure the AC losses of the HTS REBCO winding. It should be noted that we have found that due to the large inductance of the superconducting winding, it is difficult to compensate for the phase angle between the measured AC voltage signal and the AC current signal. In order to achieve better compensation for the AC voltage signal and AC current signal during the experiment, we wound a copper winding of the same size as the superconducting winding to perform the compensation. In this paper, the copper winding was placed 25 mm above the HTS REBCO winding, as shown in figure 20(c). According to our calculations, the copper winding required 50 turns.

Firstly, the critical current of the HTS REBCO winding is measured. The power amplifier (EP4000) supplies a DC current to the superconducting winding through the current leads, and the voltage of the superconducting winding is measured by the twisted voltage leads attached to both ends of the superconducting winding. In the liquid nitrogen bath environment, we measured the voltage of the

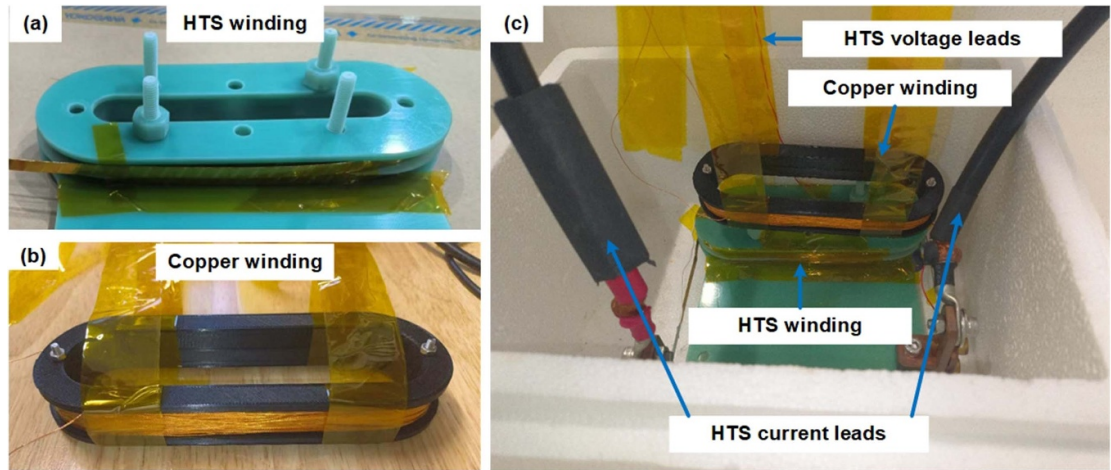


Figure 20. (a) Diagram of the HTS REBCO winding. (b) Diagram of the compensation copper winding. (c) AC loss measurement platform of the REBCO winding.

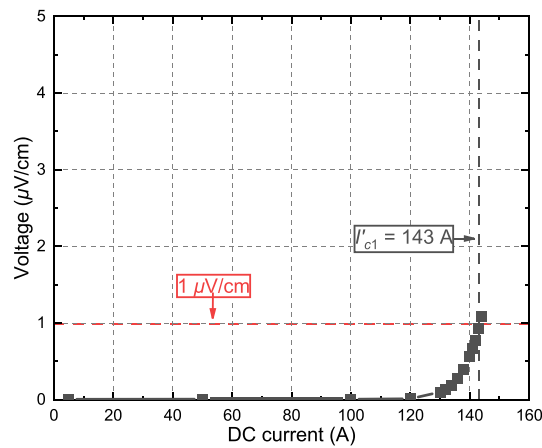


Figure 21. Measured V - I curve of the HTS REBCO winding under various DC currents. © (2026) IEEE. Reprinted, with permission, from [76].

superconducting winding under different DC currents, as shown in figure 21. The quench criterion of the superconducting winding is $1 \mu\text{V cm}^{-1}$, thus the measured critical current of the superconducting winding is 143 A.

Then, the AC transport loss of the HTS REBCO winding is measured under different AC currents and frequencies. During the experiment, a power amplifier (EP4000) and a transformer are used to supply different amplitudes and frequencies of the AC current into the HTS REBCO winding. The voltage of the HTS winding was measured simultaneously from the HTS twisted voltage leads attached to both ends of the HTS winding by the phase-locked amplifier (SR830). Then, the AC loss of the HTS REBCO winding was obtained by multiplying the AC voltage and the AC current. In this paper, the amplitude of the AC current is set from 0.1 to 0.9 I_{c0} , and the frequency is from 50 to 200 Hz. The experimental and simulation results are shown in figure 22.

It shows that the AC transport loss of the HTS REBCO winding increases with the AC current and frequency. It can be calculated that the AC transport loss of an HTS REBCO armature winding is 0.123 W at the working condition of the rated power ($I_{t,\text{peak}} = 0.2 I_{c0}$, $f = 167 \text{ Hz}$). It also can be observed that the experimental and simulation results are highly consistent, with an error of less than 10%. Therefore, the superconducting modeling method proposed in this paper and the experimental results can mutually confirm each other. Therefore, to reduce the AC transport loss, the operating AC current needs to be designed at a low level in the fully superconducting electrical machines.

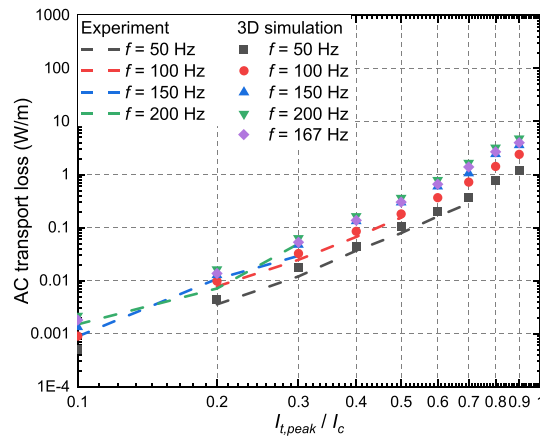


Figure 22. AC transport loss of the REBCO winding under various AC currents and frequencies. © (2026) IEEE. Reprinted, with permission, from [76].

6. Conclusion

In order to consider the unique properties of the HTS machine windings in superconducting electrical machines, we propose an efficient and accurate MVPB method in this paper. Based on this MVPB method, a 3 MW and 5000 rpm fully superconducting SM is comprehensively designed and optimized for future electric aircraft, including its dynamic loss, AC loss, cost, PWR (machine-only), and efficiency. Through this study, we identified key findings:

- (1) The MVPB method shows a high accuracy compared with the widely used fully coupled method, and the inaccuracy of the dynamic loss and AC loss is lower than 1% and 2%, respectively. Meanwhile, the MVPB method can reduce the computation time by 96%. This method is a quick and accurate method to predict the critical current and losses of the HTS machine windings at a stable operation state of the superconducting electrical machines.
- (2) For superconducting electrical machines, it is necessary to consider the superconducting properties of the HTS machine windings in the design and optimization process. By using the joint simulation (modelFRONTIER & MATLAB & COMSOL), the losses, superconductor's total length, and PWR (machine-only), and other key parameters of superconducting electrical machines can be comprehensively optimized, making the designs more reliable.
- (3) To increase the efficiency, the multifilamentary method is used to reduce the AC loss of the HTS armature windings. The results show that filamentary structure can effectively reduce AC losses of HTS windings. The efficiency of the superconducting electrical machines can be improved by approximately 8% when adopting a 4-filamentary structure HTS winding.

This work provides a powerful tool to effectively and accurately estimate the loss of the HTS machine windings. Meanwhile, it offers a good understanding of designing high power-to-weight and high-efficiency superconducting electrical machines. Additionally, the results can provide a basis for the R&D of the superconducting electrical machine prototype for future electric aircraft.

Data availability statement

All data that support the findings of this study are included within the article (and any supplementary files).

Appendix

In order to provide a quantitative estimate or reference for the error introduced by the end-winding effect, we separately compare the AC transport loss, AC magnetization loss, and total AC loss of the HTS REBCO winding between the 2D and 3D models. The 2D and 3D HTS REBCO winding models are all built using the same parameters based on T - A formulation.

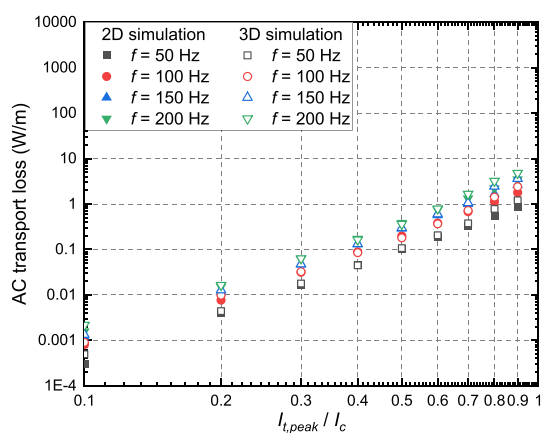


Figure 23. AC transport loss comparison of the 2D simulation and 3D simulation under different AC currents and different frequencies.

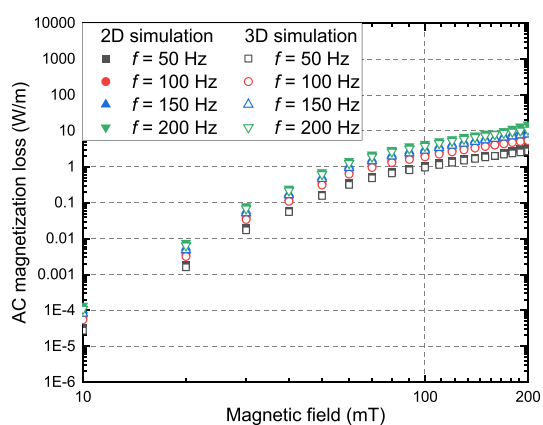


Figure 24. AC magnetization loss comparison of the 2D simulation and 3D simulation under different AC magnetic fields and different frequencies.

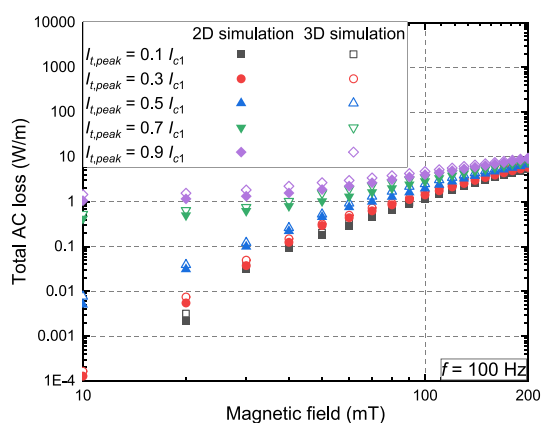


Figure 25. Total AC loss comparison of the 2D simulation and 3D simulation under different AC currents and different AC magnetic fields.

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